

BUSINESS REQUIREMENTS DOCUMENT (BRD)

Gold Coin Manufacturing, Logistics & Verified Last-Mile Delivery Platform

Tamil Nadu Government – Gold Coin Scheme (24-Carat / 8 gm Gold Coin)

Prepared For	Titan Company Limited – Program & Technology Team
Client / Sponsor	Government of Tamil Nadu – Social Welfare Department
Document Type	Business Requirements Document (Technology Platform)
Version	1.0 (Draft for Internal Review)
Date	03 July 2026
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1. Document Control

1.1 Version History

Version	Date	Author	Description of Change
1.0	03-Jul-2026	Bhushan Nirgude, Harshad Save	Detailed BRD incorporating TN Govt brief & full tech scope

1.2 Reviewers & Approvers

Name / Role	Organization	Responsibility
Program Sponsor	Titan CBG	Overall approval
Business Owner	Titan CBG	Requirement sign-off
Technology Lead	CBG- Digital	Solution feasibility
Logistics Head	Titan CBG & Tanishq	Delivery process sign-off

2. Executive Summary

The Government of Tamil Nadu (TN Govt), through its Social Welfare Department, provides 8 grams of 24-carat gold to eligible newlywed brides under its marriage assistance programs. Titan Company Limited has been engaged as the manufacturing and fulfilment partner responsible for producing the gold coins and delivering them securely to each individual beneficiary through a strictly verified last-mile process.

This BRD defines the business and functional requirements for a secure, auditable, end-to-end technology platform that manages the complete lifecycle: beneficiary data onboarding, manufacturing coordination via ERP, central and local logistics orchestration, appointment scheduling, verified last-mile delivery with a 3-level identity check, tamper-proof proof-of-delivery capture, and scheduled reporting to the TN Government.

The platform must enforce the non-negotiable rule that a gold coin is handed over ONLY to the verified beneficiary — never to a treasury, government office, or family member — even in remote (Tier-4) locations, with support for multiple delivery attempts.

3. Project Background & Context

- TN Govt distributes 24-carat, 8 gm gold coins to eligible newlywed brides under marriage assistance schemes.
- Titan is the manufacturing and fulfilment partner; TN Govt does NOT want to handle physical coins at any point.
- TN Govt will share the beneficiary master list (details + addresses + a unique beneficiary code) with Titan.
- TN Govt will provide 30 days advance notice on required coin quantity per cycle.
- TN Govt has confirmed it cannot provide a Tamil Nadu pincode master list; Titan must source this from its courier partner.
- TN Govt has requested a serviceability matrix: pincodes deliverable within 7 days and within 7–14 days.

- Delivery must follow a mandatory 3-step verification and a robust proof-of-delivery mechanism.
- Next review meeting with TN Govt is scheduled for Tuesday, 7th July.

4. Business Objectives

#	Objective	Success Indicator
O1	Digitally onboard and validate beneficiary data received from TN Govt	100% of records validated with error reporting
O2	Integrate demand with manufacturing (ERP) and track coin production	Real-time production status per cycle
O3	Orchestrate central → local logistics with pincode-based segregation	Automated routing per pincode
O4	Enable verified, appointment-based last-mile delivery	≥ 95% first/second-attempt success
O5	Enforce strict 3-level + unique-code identity verification	0 unverified handovers
O6	Capture tamper-proof proof of delivery (photo + GPS + timestamp)	100% deliveries with valid PoD
O7	Provide real-time admin dashboards and scheduled TN Govt reports	Reports on agreed frequency, on time
O8	Maintain a defensible audit trail for dispute/claim handling	Full traceability per coin

5. Scope

5.1 In Scope

- Beneficiary Data Onboarding module (bulk import, validation, de-duplication, unique-code mapping).
- ERP integration layer for demand-to-manufacturing communication and coin/production tracking.
- Inbound validation of manufactured coins and coin-to-beneficiary tagging (serialization).
- Central Logistics module with pincode-wise address repository and serviceability (7-day / 7–14 day) matrix.
- Segregation & dispatch routing from central to local logistics hubs.
- Outbound call / appointment scheduling module (1 day in advance).
- Mobile-responsive Delivery Partner application for last-mile execution.
- 3-level verification engine: Unique Code + Mobile OTP + Aadhaar-based OTP.
- Proof-of-Delivery capture: beneficiary photograph with embedded date/time + GPS coordinates/location pin.
- Real-time Admin Panel (monitoring, exceptions, re-attempts, claim/dispute management).
- Reporting & data-sharing module for TN Govt (scheduled + on-demand).
- Role-based access control, security, and complete audit logging.

5.2 Out of Scope

- Manufacturing coins (handled operationally, not by the platform beyond tracking).
- Any delivery to treasury, government offices, or third parties (explicitly prohibited by TN Govt).
- Determination of beneficiary eligibility (owned by TN Govt).
- Financial disbursement or payment processing to beneficiaries.

6. Stakeholders & User Roles

Role	Interface	Key Responsibilities
TN Govt Nodal Team	Reports / Portal (view only)	Provide beneficiary + unique codes; receive reports
Titan Backend / Program Team	Admin & Onboarding Web Portal	Receive & validate data; trigger ERP demand
Manufacturing / Factory Team	ERP Interface	Produce coins; update production status
Central Logistics Team	Central Logistics Web Portal	Validate inbound, pincode segregation, dispatch
Local Logistics Team	Local Hub Web/Mobile Portal	Receive units, schedule, assign delivery agents
Outbound Call Team	Scheduling Module	Confirm beneficiary presence & preferred slot
Last-Mile Delivery Partner	Mobile-Responsive App	Verify identity, deliver, capture PoD
Titan Admin / Control Tower	Admin Panel Dashboard	Real-time monitoring, exceptions, claims
System / Security Admin	Admin Console	User & role management, audit, config

7. Assumptions, Dependencies & Constraints

7.1 Assumptions

- TN Govt shares accurate beneficiary data including name, address, pincode, Aadhaar-linked mobile number, and unique code.
- Aadhaar-based OTP authentication is enabled via an authorized/licensed authentication channel.
- The courier/logistics partner can provide the Tamil Nadu pincode serviceability list.
- Coin quantity notice will be given 30 days in advance; volumes above 25,000/month get additional production time.
- Beneficiary mobile numbers are active and Aadhaar-registered at time of delivery.

7.2 Dependencies

- Courier partner API/data for pincode serviceability and delivery tracking.
- ERP system availability and integration endpoints.
- Authorized Aadhaar OTP authentication service (UIDAI-compliant channel).
- SMS gateway for mobile OTP and notifications; telephony for outbound calling.

7.3 Constraints

- Zero tolerance: handover only to the verified beneficiary — not family members even if they possess all codes.
- No delivery to treasury or government locations under any circumstances.
- High-value goods (gold) require strong security, insurance, and chain-of-custody controls.
- Remote areas may have weak mobile signal/transport — offline capability required.
- Data privacy compliance for Aadhaar and personal data (India DPDP Act).

8. Solution Overview & High-Level Architecture

The platform is a modular, cloud-based system comprising role-specific web and mobile interfaces sitting on a common secure data backbone. Data flows sequentially through nine stages, with a real-

time control-tower (Admin Panel) providing oversight and a reporting layer serving the TN Government.

End-to-End Data Flow:

[1] TN Govt Beneficiary Data + Unique Codes → [2] Titan Backend Validation & Onboarding → [3] ERP Demand to Factory → [4] Coin Production & Serialization → [5] Inbound Validation & Coin-Beneficiary Tagging → [6] Central Logistics (Pincode Repository + Serviceability Matrix + Segregation) → [7] Dispatch to Local Logistics Hub → [8] Outbound Call / Appointment Scheduling (T-1 day) → [9] Last-Mile Delivery App: 3-Level Verification + PoD Capture → [10] Real-Time Admin Panel → [11] Scheduled Reporting to TN Govt

9. Functional Requirements

9.1 Module 1 – Beneficiary Data Onboarding (Backend Team)

Interface for the Titan backend team to securely receive, validate, and manage beneficiary data from TN Govt, and to communicate demand to the factory through the ERP system.

Req ID	Requirement	Priority
FR-1.1	Support secure bulk upload / integration of beneficiary data from TN Govt (name, address, pincode, Aadhaar-linked mobile, unique code).	Must
FR-1.2	Validate mandatory fields, address format, pincode validity, and mobile number format; generate an exception/error report for rejected records.	Must
FR-1.3	De-duplicate records and detect conflicts; flag duplicates for manual review.	Must
FR-1.4	Map and store the TN Govt unique beneficiary code against each record.	Must
FR-1.5	Aggregate validated demand and transmit coin production requirement to the ERP system.	Must
FR-1.6	Maintain 30-day advance notice tracking and flag cycles exceeding 25,000 units/month for extended production timelines.	Must
FR-1.7	Maintain full status of each beneficiary record (Received → Validated → In Production → Ready → Dispatched → Delivered/Failed).	Must

9.2 Module 2 – Manufacturing / ERP Integration & Inbound Validation

Once coins are produced, they must be validated and forwarded to the central logistics team with beneficiary details.

Req ID	Requirement	Priority
FR-2.1	Receive production status updates from ERP (planned, in-progress, completed quantities per cycle).	Must
FR-2.2	Support serialization / unique identifier for each coin for chain-of-custody tracking.	Must
FR-2.3	Perform inbound validation (quantity, quality-check status, purity/weight confirmation flags) on receipt from manufacturing.	Must
FR-2.4	Tag / map each validated coin to a specific beneficiary record and unique code.	Should
FR-2.5	Forward validated units with beneficiary details and addresses to the Central Logistics module.	Must
FR-2.6	Raise alerts for shortfalls, mismatches, or rejected units.	Should

9.3 Module 3 – Central Logistics & Pincode Serviceability

Central logistics maintains a pincode-wise address repository and performs segregation of units by delivery region.

Req ID	Requirement	Priority
FR-3.1	Maintain a repository of all Tamil Nadu pincodes (sourced from courier partner) with serviceability data.	Must
FR-3.2	Classify each pincode by delivery SLA: 'Deliverable within 7 days', 'within 7–14 days', or 'beyond/manual'.	Must
FR-3.3	Generate the TN Govt serviceability matrix report (7-day and 7–14-day pincode lists).	Must
FR-3.4	Automatically segregate/route units by pincode to the correct local logistics hub.	Must
FR-3.5	Flag remote / Tier-4 / low-connectivity pincodes for special handling (offline mode, extra attempts).	Must
FR-3.6	Generate dispatch manifests (unit ID + beneficiary + address + hub) with chain-of-custody handoff records.	Must

9.4 Module 4 – Local Logistics & Appointment Scheduling

Local logistics teams receive units, and an outbound call is made one day in advance to confirm beneficiary presence and preferred slot.

Req ID	Requirement	Priority
FR-4.1	Receive units and beneficiary details from central logistics with acknowledgement/handoff confirmation.	Must
FR-4.2	Provide a scheduling interface for the outbound call team to log calls made T-1 day in advance.	Must
FR-4.3	Capture beneficiary-confirmed delivery date and preferred time slot (within working hours).	Must
FR-4.4	Record call outcomes (confirmed / no-answer / reschedule / unreachable) and trigger retry logic.	Must
FR-4.5	Assign confirmed deliveries to specific last-mile delivery agents by route/area.	Must
FR-4.6	Support rescheduling and multiple attempt tracking for unreachable beneficiaries.	Must

9.5 Module 5 – Last-Mile Delivery Partner App (Mobile-Responsive)

A mobile-responsive interface for delivery agents to execute verified handovers with proof of delivery.

Req ID	Requirement	Priority
FR-5.1	Display assigned deliveries with beneficiary details, address, appointment slot, and route.	Must
FR-5.2	Enforce the mandatory 3-level verification workflow before allowing handover (see Section 10).	Must
FR-5.3	Verify the TN Govt unique code by matching against the stored record.	Must
FR-5.4	Trigger and validate OTP sent to the Aadhaar-registered mobile number.	Must
FR-5.5	Trigger and validate Aadhaar-based OTP authentication.	Must
FR-5.6	Capture a photograph of the beneficiary at handover with embedded date, time, and GPS coordinates.	Must
FR-5.7	Drop/capture a location pin (GPS) confirming the delivery location.	Must
FR-5.8	Block handover if any verification fails; require handover to the beneficiary only (not family members).	Must
FR-5.9	Support offline capture in low-connectivity areas with auto-sync once connectivity is	Must

	restored.	
FR-5.10	Record failed attempts with reason codes and enable re-attempt scheduling.	Must
FR-5.11	Provide navigation assistance to the beneficiary address.	Could

9.6 Module 6 – Admin Panel / Control Tower

Real-time monitoring and management interface for the Titan admin team.

Req ID	Requirement	Priority
FR-6.1	Provide real-time dashboards for production, dispatch, in-transit, delivered, failed, and pending units.	Must
FR-6.2	Display live proof-of-delivery records (photo, GPS, timestamp, verification status).	Must
FR-6.3	Monitor SLA adherence (7-day / 7–14-day) with breach alerts.	Must
FR-6.4	Manage exceptions, re-attempts, escalations, and remote-location handling.	Must
FR-6.5	Provide a claim / dispute management workflow with timestamped evidence.	Must
FR-6.6	Role-based access with drill-down from state → district → pincode → beneficiary.	Should

9.7 Module 7 – Reporting & TN Govt Data Sharing

Scheduled and on-demand reporting to the TN Government at an agreed frequency.

Req ID	Requirement	Priority
FR-7.1	Generate scheduled delivery reports (daily/weekly/agreed frequency) for TN Govt.	Must
FR-7.2	Include proof-of-delivery evidence (photo, GPS, timestamp, verification confirmation) per beneficiary.	Must
FR-7.3	Provide serviceability matrix reports (pincode-wise SLA classification).	Must
FR-7.4	Support secure export/sharing (portal access, secure file, or API) with access controls.	Must
FR-7.5	Provide reconciliation reports (units demanded vs produced vs delivered vs pending).	Must
FR-7.6	Maintain historical/audit reports for dispute resolution.	Should

10. Three-Level Verification & Proof of Delivery (Critical Control)

The handover of a coin must be blocked unless ALL of the following are successfully completed, in sequence:

Step	Verification	System Behaviour
Pre-Step	TN Govt Unique Code match	Agent enters/scans unique code; system matches against stored beneficiary record.
Level 1	Mobile OTP	OTP sent to Aadhaar-registered mobile number; must be entered & validated.
Level 2	Aadhaar-based OTP	Aadhaar OTP authentication performed & validated via authorized channel.
Level 3	Photograph + GPS + Timestamp	Photo of beneficiary captured with embedded date, time, and GPS coordinates; location pin dropped.

Business Rules:

- Handover is permitted ONLY to the verified beneficiary — never to a family member, even if they possess the OTP, unique code, and Aadhaar OTP.
- If any verification step fails, the app must prevent handover and log a failed attempt with a reason.

- All verification artefacts (codes matched, OTP status, photo, GPS, timestamp) are stored as an immutable audit record.
- No delivery to treasury or government office is permitted under any condition.

11. Non-Functional Requirements

Category	Requirement
Security	End-to-end encryption; role-based access; secure Aadhaar handling; audit logging of every action.
Data Privacy	Compliance with India DPDP Act; consent, data minimization, and secure Aadhaar OTP handling.
Availability	≥ 99.5% uptime for core services; graceful degradation.
Offline Capability	Delivery app must operate offline in low-signal areas and auto-sync on reconnection.
Scalability	Support high-volume cycles (>25,000 units/month) and peak reporting loads.
Performance	Verification steps and PoD upload complete within acceptable field-response times.
Auditability	Immutable, timestamped chain-of-custody and proof records per coin.
Usability	Mobile-responsive, multilingual (Tamil/English) interfaces for field usability.
Reliability	Idempotent sync, retry logic, and no data loss during connectivity gaps.
Integrations	ERP, courier partner, SMS gateway, Aadhaar OTP channel, telephony.

12. Integration Requirements

- ERP System – demand transmission and production status updates.
- Courier / Logistics Partner – pincode serviceability data and shipment tracking.
- Aadhaar OTP Authentication – via an authorized/licensed UIDAI-compliant channel.
- SMS Gateway – mobile OTP and beneficiary notifications.
- Telephony / Call System – outbound call scheduling (T-1 day).
- TN Govt Reporting Channel – secure portal / file / API for report sharing.

13. Key Data Requirements

Data Entity	Key Attributes
Beneficiary	Name, address, pincode, district, Aadhaar-linked mobile, TN Govt unique code, status
Coin / Unit	Serial ID, purity/weight, production cycle, mapped beneficiary, custody status
Pincode Repository	Pincode, region, hub, serviceability SLA (7-day / 7–14-day / manual)
Appointment	Confirmed date, preferred slot, call outcome, assigned agent
Delivery / PoD	Unique code match, mobile OTP status, Aadhaar OTP status, photo, GPS, timestamp, attempt no., result
Claim / Dispute	Claim date, beneficiary, reason, evidence, resolution, SLA (2–3 day rule)

14. Risks & Mitigation

Risk	Impact	Mitigation
Weak mobile signal in remote villages	OTP/verification failure	Offline capture + auto-sync; alternate verification protocol to be agreed with TN Govt
Poor transport access to Tier-4 areas	Delivery delays	Special routing, extended SLA, multiple attempts
Aadhaar OTP channel dependency	Verification blocked	Use authorized channel; define fallback/escalation

		with TN Govt
High-value goods security	Theft/loss risk	Serialized chain-of-custody, insurance, verified handover only
Beneficiary unreachable	Repeated failed attempts	T-1 outbound call, rescheduling, defined max attempts
Non-delivery claims	Disputes	Immutable PoD evidence; propose 2–3 day claim window (pending TN Govt confirmation)
Volume spikes >25,000/month	Production strain	Leverage 30-day notice + agreed extended timelines

15. Service Levels & Delivery Rules

- Serviceability classification: pincodes deliverable within 7 days and within 7–14 days, sourced from courier partner.
- Outbound confirmation call to be made 1 day (T-1) prior to delivery.
- Delivery only during beneficiary's confirmed working-hours slot.
- Multiple delivery attempts mandatory for remote/unreachable beneficiaries.
- Proposed claim/dispute window: 2–3 days post-delivery (TN Govt to confirm).
- 30-day advance quantity notice; extended production time for cycles above 25,000 units/month.

16. Open Items / Points to Raise with TN Govt (Meeting: 7 July)

#	Open Point	Why It Matters
1	Confirmation of the 2–3 day claim/non-delivery window	Needed to close dispute-handling SLA
2	Agreed fallback verification for zero-signal remote areas	Aadhaar/mobile OTP may be impossible offline
3	Maximum number of delivery re-attempts before return protocol	Defines cost & operational limits (no return to treasury)
4	Format, frequency & channel for TN Govt reports	Drives reporting module design
5	Provisioning of authorized Aadhaar OTP authentication channel	Regulatory/technical enablement
6	Data-sharing security & privacy expectations (DPDP compliance)	Aadhaar & PII handling
7	Confirmation that courier partner pincode list is acceptable as source of truth	TN Govt cannot supply pincode master
8	Handling of undeliverable coins (since no treasury/govt returns allowed)	Needs a defined exception path

17. Success Metrics (KPIs)

- % deliveries completed within committed SLA (7-day / 7–14-day).
- First-attempt and overall delivery success rate.
- % deliveries with complete & valid 3-level verification + PoD.
- Average delivery turnaround time by pincode tier.
- Number of exceptions, re-attempts, and disputes (with resolution time).
- Reconciliation accuracy: demanded vs produced vs delivered.
- On-time submission rate of TN Govt reports.

18. Glossary

Term	Definition
Beneficiary	Eligible newlywed bride entitled to the gold coin
Unique Code	TN Govt-issued code mapped to each beneficiary, verified at delivery
PoD	Proof of Delivery – photo + GPS + timestamp + verification record
OTP	One-Time Password
SLA	Service Level Agreement (delivery timelines)
Tier-4 / Remote	Low-connectivity, hard-to-access rural locations
Control Tower	Real-time admin monitoring & exception management panel

19. Sign-Off

Role	Name	Signature	Date
Business Owner (Titan)	Rahul Pandey		
Technology Head	Srinivasan Krishnaraj		
Project Head	Vikram Bengre		
Program Managers	Bhushan Nirgude & Harshad Save		